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Ph.D in Physics, working with computational physics/chemistry and developing scientific software. My research interests are in the development of new methods for molecular simulation and the use of such methods to solve problems in physical chemistry, biochemistry, and materials science.

Education and positions

2024– **Researcher, Hylleraas Centre for Quantum Molecular Sciences, University of Oslo**
Working in the H₂O-MOF project developing new methods and applying machine learning potentials to investigate the conductivity of ionic species through metal-organic frameworks membranes.

2022–2024 **Postdoctoral researcher, Hylleraas Centre for Quantum Molecular Sciences, University of Oslo**
Further developments in the Hamiltonian hybrid particle-field molecular dynamics and use with metainference - Supervisor: Prof. Michele Cascella

2020–2022 **Postdoctoral researcher, Institute of Physics, University of São Paulo**
Computational nanofluidics studies on the confinement effects on fluid-solid interfaces.
FAPESP Fellowship - Supervisor: Prof. Caetano R. Miranda

2018–2019 **Postdoctoral researcher, Institute of Physics, University of São Paulo**
Development and applications of Configurational Bias Monte Carlo Method in dyes of interest to organic solar cells
CAPES Fellowship - Supervisor: Prof. Kaline R. Coutinho

2015–2018 **Ph.D. in Physics, Institute of Physics, University of São Paulo**
Implementation and Development of Efficient Algorithm for Intramolecular Deformation with the Monte Carlo Method
CNPq Ph.D. Fellowship - Advisor: Prof. Kaline R. Coutinho

2013–2015 **Master's in Computational Physics, São Carlos Institute of Physics, University of São Paulo**
Implementation of the Parallel Tempering Monte Carlo Method to the Study of Thermodynamic Properties of Nanoclusters
FAPESP Master Fellowship - Advisor: Prof. Juarez L. F. Da Silva

2009–2012 **B.S. in Computational Physics, São Carlos Institute of Physics, University of São Paulo**
Undergraduate research entitled: Fluid Dynamics in Porous Media with Lattice Boltzmann
CNPq Scientific Initiation Fellowship - Advisor: Prof. Leonardo P. Maia

Approved grants

2020 **FAPESP Two-years Postdoc Fellowship.**
2013 **FAPESP Two-years Masters Fellowship.**

Awards

2022 **Best PhD thesis** in the field of atomic and molecular physics, by the Brazilian Physical Society.

- 2022 **Honorable mention** at the annual prize of the Institute of Physics, University of São Paulo, at the “Highlight in instrumentation and software development” category.
- 2020 **Top Downloaded Paper**: “Solvent effect on the syn/anti conformational stability: A comparison between conformational bias Monte Carlo and molecular dynamics methods”, Wiley.
- 2019 **3rd place** in the best flash talks award at the XX Simpósio Brasileiro de Química Teórica.
- 2017 **Best poster** award at the XIX Simpósio Brasileiro de Química Teórica.
- 2017 **Among the 12 finalists** of the “My Thesis in 180 Seconds”, organized by Swissnex Brazil.
- 2015 **Honorable mention** at the best poster award of the II Workshop on Biomolecular Theory-Experiment Interplay.

Teaching Experience

- 2019 **Física II (Basic Physics II)**: Teacher for the second semester physics class for the mechanical engineering majors at the Escola Politécnica – University of São Paulo.
- 2019 **Física I (Basic Physics I)**: Teacher for the first semester physics class for the mechanical engineering majors at the Escola Politécnica – University of São Paulo.
- 2017 **Física Moderna I (Modern Physics I)**: Teaching assistant for the introductory modern physics class for the physics majors of the Institute of Physics – University of São Paulo, under the supervision of Prof. Dr. Sylvio Canuto.

Research Experience

My research is mainly focused on the development of algorithms for classical molecular simulation. Specifically, I have experience implementing and using Monte Carlo methods and molecular dynamics for the simulation of solute-solvent, interfacial and biomolecular systems. I also have experience using Metainference to integrate experimental and simulation data, and have given contributions to PLUMED in this regard. On top of the molecular simulation methods, I have some experience with global optimization algorithms, clustering algorithms, fitting of interatomic potentials, and density functional theory. Recently, my collaborators and I have been working on the development of a active learning method to train machine learning potentials for molecular simulations using less reference data. The method is called Calibrated Adversarial Geometry Optimization (CAGO) and generates new configurations based on high uncertainty forces.

Languages

Portuguese (native), English (fluent) and Norwegian (novice).

Software development

[GitHub](#), [GitLab](#)

Apart from many scripts and smaller software (see the links above), my major contributions are developing:

Cezar HM, Canuto S, Coutinho K

DICE

<https://portal.if.usp.br/dice>

A Monte Carlo code for the molecular simulation of liquids and gases using classical force fields. My main contribution is the coding and development of a Configurational Bias Monte Carlo method used to sample the internal degrees of freedom of the molecules, and parallelization. The code is written in Fortran and has OpenMP parallelization.

Rondina GG, Cezar HM, Da Silva JLF

GOTNano

A code for global optimization and sampling of thermodynamic properties of nanoclusters. My main contribution is the implementation of the Parallel Tempering Monte Carlo algorithm used to sample the thermodynamic properties. The code is written in C and has OpenMP parallelization.

Ledum M, Carrer M, Sen S, Li X, Cezar HM, Cascella M, Bore SL

Hylleraas MD

<https://github.com/Cascella-Group-UiO/HyMD>

A molecular dynamics software that implements the Hamiltonian hybrid particle-field method. I am currently the main maintainer of the software, and my main contribution is the PLUMED interface. HyMD is written in Python and Fortran and uses MPI for parallelization.

Cezar HM, Ribeiro RB

ClustTraj

<https://github.com/hmcezar/clusttraj>

A parallel Python package that performs clustering over an atomistic molecular simulation trajectory, searching for the minimum RMSD between each structure and classifying using hierarchical clustering, with options to deal with solute solvent systems.

Cezar HM et al.

Hylleraas Software Platform

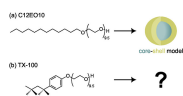
<https://gitlab.com/hylleraasplatform>

A platform to facilitate creating workflows and working with scientific software. My main contributions are to the development of the active learning package, and the interface for the training and evaluation of NequIP/Allegro models.

Publications

[Web of Science](#), [Scopus](#), [ORCID](#), [Google Scholar](#)

Published

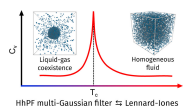


Cezar HM, Bjørnstad VA, Prévost S, Lund R, Cascella M

Beyond Core-Shell Micellar Structures: Complex Structures in Simple Surfactants

Small Structures 2400553, 2024 · [DOI](#)

We show how using Metainference one can go beyond fitting simple geometric models to interpret SAXS and SANS data, and how this is necessary even for relatively simple systems. We also address Triton X-100 aggregates, for which structures have been debated.

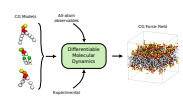


Sen S, Cezar HM, Ledum M, Li X, Cascella M

Phase coexistence in Hamiltonian hybrid particle-field theory using a Multi-Gaussian approach

The Journal of Physical Chemistry B 128:11739, 2024 · [DOI](#)

This study introduces an implementation of multiple Gaussian filters within the Hamiltonian hybrid particle-field (HhPF) theory. Using a two Gaussian filter we show that HhPF effectively captures detailed information on phase coexistence and interfacial phenomena.

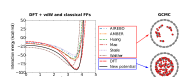


Carrer M, Cezar HM, Bore SL, Ledum M, Cascella M

Learning Force Field Parameters from Differentiable Particle-Field Molecular Dynamics

Journal of Chemical Information and Modeling 64:5510, 2024 · [DOI](#)

We develop a differentiable molecular dynamics software based on the Hamiltonian hybrid particle-field formalism, and use it to establish a protocol for automated optimization of force field parameters.

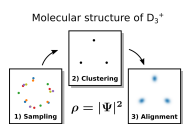


Cezar HM, Lanna TD, Damasceno DA, Kirch A, Miranda CR

Revisiting greenhouse gases adsorption in carbon nanostructures: advances through a combined first-principles and molecular simulation approach

Applied Surface Science 671:160659, 2024 · [DOI](#)

We investigate the adsorption properties of CH₄ and CO₂ in carbon nanotubes (CNTs) using different force fields, showing that the adsorption energies are underestimated. A new interatomic potential is proposed based on the first-principles calculations.

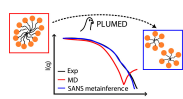


Lang L, Cezar HM, Adamowicz L, Pedersen TB

Quantum Definition of Molecular Structure

Journal of the American Chemical Society 146:1760, 2024 · [DOI](#)

In this manuscript we use Monte Carlo to sample an all-particle wavefunction, and propose a new definition of molecular structure beyond the Born-Oppenheimer approximation.

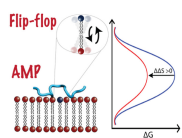


Cezar HM, Cascella M

SANS Spectra with PLUMED: Implementation and Application to Metainference

Journal of Chemical Information and Modeling 63:4979, 2023 · [DOI](#)

In this Application Note we describe our implementation of Small Angle Neutron Scattering spectra including the use of a resolution function and its application to metainference simulations.

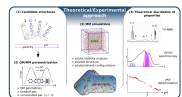


Carrer M, Nielsen JE, Cezar HM, Lund R, Cascella M, Soares TA

Accelerating Lipid Flip-Flop at Low Concentrations: A General Mechanism for Membrane Binding Peptides

The Journal of Physical Chemistry Letters 14:7014, 2023 · DOI

We show that the flip-flop of lipids in a membrane can be accelerated by the presence of peptides. The mechanism is entropic, due to the restriction in mobility of lipids introduced by the presence of the peptide, and is not necessarily driven by smaller enthalpic barriers.

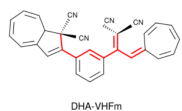


Franco LR, Toledo KCF, Matias TA, Benavides PA, Cezar HM, Araujo CMG, Coutinho K, Araki K

Unraveling the acid-base characterization and solvent effects on structural and electronic properties of a bis-bidentated bridging ligand

Physical Chemistry Chemical Physics 24:10222, 2022 · DOI

We report a combined theoretical-experimental study of the bis(2-pyridyl)-bis-imidazole in solution, determining the protonation sites, most favorable conformations and absorption spectra in different solvents.



Cardenuto MH, Cezar HM, Mikkelsen KV, Sauer SPA, Coutinho K, Canuto S

A QM/MM study of the conformation stability and electronic structure of the photochromic switches derivatives of DHA/VHF in acetonitrile solution

Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy 251:119434, 2021 · DOI

The absorption spectra of photoswitches composed of two units of photochromic molecules, namely, dihydroazulene (DHA)/vinylheptafulvene(VHF) is studied including explicit solvation and considering the rotamers that are exhibited at room temperature. We obtain the free energies and perform clustering analysis, rationalizing the absorption spectra at the different solvents.

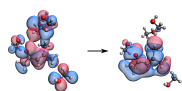


Cezar HM, Canuto S, Coutinho K

DICE: A Monte Carlo code for molecular simulation including Configuration Bias Monte Carlo method

Journal of Chemical Information and Modeling 60:3472, 2020 · DOI

We describe DICE, a software that uses Monte Carlo methods to perform molecular simulation, focusing on solute-solvent systems. We introduce our implementation of the Configurational Bias Monte Carlo method that includes new methodological developments and employ the method to study different molecules such as alkanes, 1,2-dichloroethane and a large boron phthalocyanine.

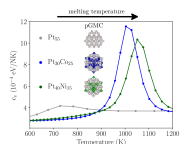


Cezar HM, Canuto S, Coutinho K

Understanding the Absorption Spectrum of Mesityl Oxide Dye in Solvents of Different Polarities

Journal of Molecular Liquids 307:112924, 2020 · DOI

We investigate the conformational and solvent effects on the UV/vis spectroscopy of the mesityl oxide dye. We show that a correct description of the syn/anti population and the use of explicit solvent molecules is fundamental to describe the solvatochromic shifts.



Cezar HM, Rondina GG, Da Silva JLF

Thermodynamic Properties of 55-Atom Pt-based Nanoalloys: Phase Changes and Structural Effects on the Electronic Properties

Journal of Chemical Physics 151:204301, 2019 · DOI

We investigate the effects of temperature on the structure of Pt-based nanoalloys, performing Parallel Tempering Monte Carlo simulations using many-body potentials and DFT calculations to see how the structural changes affect the density of states.

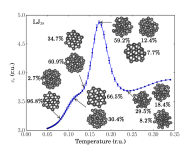


Cezar HM, Canuto S, Coutinho K

Solvent effect on the syn/anti conformational stability: A comparison between conformational bias Monte Carlo and molecular dynamics methods

International Journal of Quantum Chemistry 119:e25688, 2019 · DOI

We compare the Configurational Bias Monte Carlo (CBMC) and molecular dynamics (MD) sampling of the mesityloxide molecule in gas phase and solution, showing that while CBMC performs an ergodic sampling, MD has troubles overcoming the 10 kcal/mol energy barrier between the syn and anti conformations.

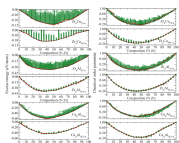


Cezar HM, Rondina GG, Da Silva JLF

Parallel Tempering Monte Carlo Combined with Clustering Euclidean Metrics Analysis to Study the Thermodynamic Stability of Lennard-Jones Nanoclusters

Journal of Chemical Physics 146:064114, 2017 · DOI

We use a clustering algorithm to analyze trajectories from Parallel Tempering Monte Carlo simulations of Lennard-Jones nanoclusters, identifying the phase changes and most frequent structures with great accuracy using a very simple approach.



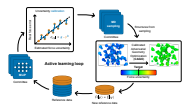
De Souza DG, Cezar HM, Rondina GG, De Oliveira MF, Da Silva JLF

A Basin-hopping Monte Carlo Investigation of the Structural and Energetic Properties of 55- and 561-atom Bimetallic Nanoclusters: the Examples of the ZrCu, ZrAl, and CuAl Systems

Journal of Physics: Condensed Matter 28:175302, 2016 · DOI

We report a basin-hopping Monte Carlo investigation of the structural and energetic properties of bimetallic ZrCu, ZrAl, and CuAl nanoclusters with 55 and 561 atoms, showing how the atoms of the different species are distributed in the structure, and a trend of more spherical structures at 50-50 compositions.

Preprints

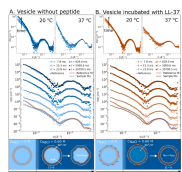


Cezar HM, Bodenstein T, Sveinsson HA, Ledum M, Reine S, Bore SL

Learning atomic forces from uncertainty-calibrated adversarial attacks

arXiv · DOI

We develop Calibrated Adversarial Geometry Optimization (CAGO). The method calibrates the uncertainty estimates for the predicted forces and generates new configurations with high uncertainty which are fed to the active learning loop.

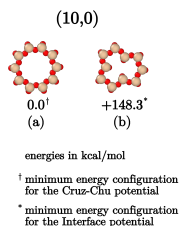


Koynarev VR, Nader ML, Borgos KKA, Cezar HM, Porcar L, Narayanan T, Cascella M, Lund R

Beyond Structural Pores: Transient Permeabilization of Lipid Membranes by Antimicrobial Peptides

ChemRxiv · DOI

Theory and experiments are combined to reveal that even unstructured and non-penetrating antimicrobial peptides (AMPs) can rapidly permeabilize lipid membranes, with complete ion equilibration occurring in only a few tens of milliseconds.

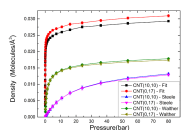


Cezar HM, Miranda CR

Water adsorption in ultrathin silica nanotubes

arXiv · DOI

We investigate the structure and adsorption properties for three sizes of SiO₂ nanotubes, understanding the adsorption in terms of the accessible volume.

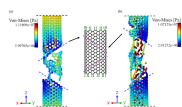


Martins Junior CA, Cezar HM, Damasceno DA, Miranda CR

Effects of van der Waals interaction on the N₂ adsorption on carbon nanotubes: proposal of new force field parameters

arXiv · DOI

We investigate the adsorption of N₂ in CNTs, proposing an interatomic potential that correctly describe the energies for different adsorption sites, leading to a more accurate description of adsorption isotherms.



Damasceno DA, Cezar HM, Lanna TD, Kirch A, Miranda CR

Mechanical and adsorption properties of greenhouse gases filled carbon nanotubes

arXiv · DOI

Using molecular simulations we investigate the mechanical stability of CNTs filled with mixtures of CO₂ and CH₄, finding that under compression several parameters can play a role in the materials response.

Selected invited and contributed talks

Dublin, Ireland
Jul 2024

Metainference simulations to interpret small angle scattering experiments

9th EuChemS Chemistry Congress

São Carlos, Brazil
Sep 2019

DICEtools and Clustering-Traj: From simulation setup to data analysis

III New Energies Innovation Center - Computational Material Science Division Workshop

São Carlos, Brazil
Feb 2019

Monte Carlo Methods

II New Energies Innovation Center - Computational Material Science Division Workshop (Invited talk)

Brasília, Brazil
Sep 2018

Comparison Between Configurational Bias Monte Carlo and Molecular Dynamics Sampling for Sampling

Conformational Stability with High Energy Barriers
VII SeedMol

Livorno, Italy
Aug 2017

Configurational Bias Monte Carlo of Molecules in Solvent and Comparison with Molecular Dynamics
Coding Solvation Workshop

Rio de Janeiro, Brazil
Mar 2017

How Indiana Jones is actually a bad explorer when it comes to molecules and how I can do better
Finals of My Thesis in 180 seconds (Swissnex Brazil)

João Pessoa, Brazil
Sep 2014

Parallel Tempering Monte Carlo Investigation of Phase-Changes in Nanoclusters
XIII Brazilian Materials Research Society Meeting

Conferences organized

Recife, Brazil
May 2017

III Advanced School on Biomolecular Simulation
Member of the local committee.